Logical Transductions for Tonal Patterns Triggered by Vowel Deletion

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1 Introduction

Vowel hiatus is avoided in many languages through processes such as vowel deletion or modification. In tonal languages, where vowels typically serve as tone-bearing units (TBUs), hiatus resolution often triggers changes in the association between tones and vowels. These include tone deletion, contour formation, and tone coalescence. Such patterns that linear phonology has been struggled to capture have been extensively studied in autosegmental phonology. From a autosegmental perspective, the tonal processes can be generalized by modifying the association relations between tones and vowels, which shows simpler generaiztions than changing the tonal or vocalic features themselves. This also demonstrates the distinct natures of tonal and segmental phonology. In the previous computational phonology work, little work has been done to model tonal processes over autosegmental representations. This paper proposes that most of tonal processes can be computationally modeled by two logical transduction: reassociation and deassociation. We show that these two operations can capture the three typologically common outcomes of vowel deletion, as well as less common patterns such as vowel deletion blocking. We further show how these processes can be learned using a structured hypothesis space, drawing connections to recent work on computational learning in phonology.

2 Typology

We argue that three typologically common outcomes of vowel deletion can be derived from two basic operations on association relations: *deassociation* and *reassociation*. Specifically:

1. Tone deletion arises from deassociation without subsequent reassociation.
2. Contour formation arises from reassociation of a tone to an already-associated TBU.
3. Tone coalescence arises from reassociation together with constraints on tonal sequences.

We further show that this system also applies to less common patterns. In particular, it captures cases of tone coalescence in Moba and Yoruba, as well as vowel deletion blocking in Ezinaihite Mbaise, where association relations constrain the applicability of the deletion process.

We formalize these processes as logical transductions and show how they operate over autosegmental representations with tone, vowel, and association relations. We then illustrate the analysis with case studies from typologically diverse languages. Finally, we outline how such processes can be learned using a structured hypothesis space, drawing connections to recent work on computational learning in phonology.

Process	Condition	Language
Tone Deletion	Tone quality Association faithfulness Morphosyntax	
Tone Retention	Contour Formation Tone Coalescence	Ewe Moba, Yoruba

Table 1: Typology of tonal processes triggered by vowel deletion

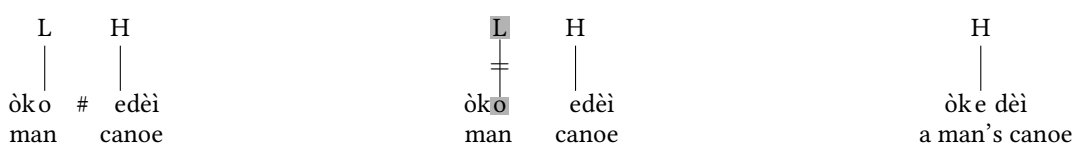
3 Logical Transduction

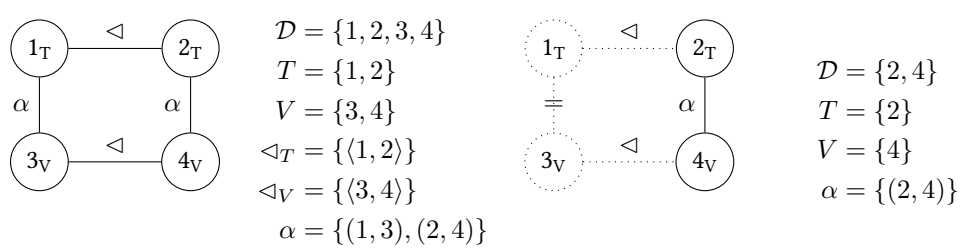
Yolyan (2025) provides a general framework for modeling phonological maps over strings. The input and output strings are represented as relational structures, and the transduction is defined by a set of logical formulas that specify how the output structure is derived from the input structure. We follow Yolyan's formalization and apply to AR using the common patterns.

3.1 Tone Elision

In languages that the vowel associated with the deleting vowels will also get deleted, this can be solved by a simple deassociation operation. The unassociated tone will stay unassociated without any further operation, which will not be realized in the surface form.

An example can be drawn from Engenni (Niger-Congo, Nigeria). When two vowels are adjacent, the first vowel always get deleted along with its associated tone.





A transduction for tone deletion can be defined as follows:

$$
\begin{aligned}
H'(x) :=& H(x) \\
L'(x) :=& L(x) \\
V'(x) :=& V(x) \wedge \neg \mathsf{delete_V}(x) \\
\lhd'_T(x,y) :=& \lhd_T (x,y) \\
\lhd'_V(x,y) :=& \lhd_V (x,y) \\
\alpha'(x,y) :=& \alpha(x,y) \wedge \neg \mathsf{deassoc}\ (x,y) \text{ where} \\
\mathsf{deassoc}(T,V) :=& \alpha(T,V) \wedge \mathsf{delete_V}(V)
\end{aligned}
$$

It should be mention that the deletion condition is language specific. For the example, Engenni deletes the proceeding vowel in the hiatus and in other language the following. These two conditions can be captured by the following formulas respectively:

Engenni: $\mathsf{delete_V}(V) := \exists V' s.t \lhd (V', V)$
L2: $\mathsf{delete_V}(V) := \exists V' s.t \lhd (V, V')$
L3: $\mathsf{delete_V}(V) := \exists V s.ta(V) \wedge \lhd(V', V) \vee \lhd(V, V')$

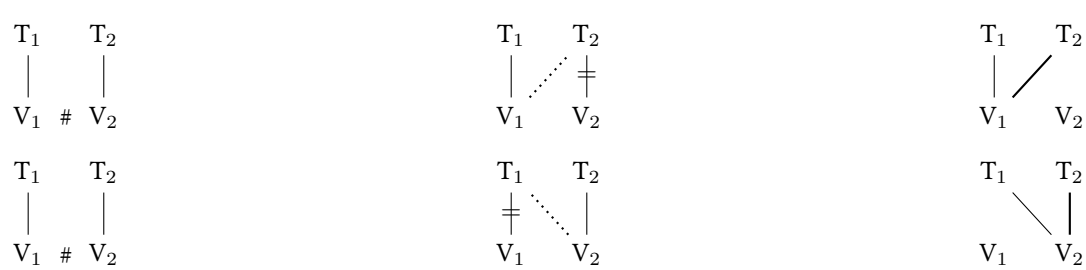
As can be seen from the tranduction, everything stays the same except the association relation. The deassociation and reassociation conditions are what determine the output association relation. For the contour formation

$$
\begin{aligned}
\mathsf{deassoc}(T,V) :=& \alpha(T,V) \wedge \exists V' \mathsf{st} \lhd (V', V) \wedge \neg \alpha(T, V') \\
\mathsf{reassoc}(T,V) :=& \neg \alpha(T,V) \wedge \exists V' \mathsf{st} \lhd (V, V') \wedge \alpha(T, V')
\end{aligned}
$$

3.2 Contour Formation

Vowel deletion is a common remedy for vowel hiatus, which often further results in tonal instability. As a result, the tone associated with the deleted vowel may disassociate from the deleted vowel and reassociate to the surviving vowel, resulting in a contour formation. The contour usuallay follows the linear order of two tones. Based on Hao, all languages in the survey but not reversed order. The logical transduction for contour formation is as follows:

Contour tone formation in Etsako



$$
\begin{aligned}
H'(x) :=& H(x) \\
L'(x) :=& L(x) \\
V'(x) :=& V(x) \\
\lhd'_T(x,y) :=& \lhd_T (x,y) \\
\lhd'_V(x,y) :=& \lhd_V (x,y) \\
\alpha'(T,V) :=& (\alpha(T,V) \wedge \neg \mathsf{deassoc}(T,V)) \vee \mathsf{reassoc}(T,V)
\end{aligned}
$$

As can be seen from the tranduction, everything stays the same except the association relation. The deassociation and reassociation conditions are what determine the output association relation.

For the contour formation

$$
\begin{aligned}
\mathsf{deassoc}(T,V) :=& \alpha(T,V) \wedge \exists V' \mathsf{st} \lhd (V',V) \wedge \neg\alpha(T,V') \\
\mathsf{reassoc}(T,V) :=& \neg\alpha(T,V) \wedge \exists V' \mathsf{st} \lhd (V,V') \wedge \alpha(T,V')
\end{aligned}
$$

References

Yolyan, T. (2025). Logical transductions for autosegmental representations. In Proceedings of the Workshop on Models of Language (MOL), Online. Association for Computational Linguistics.